

PERSONAL INFORMATION

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Nationality	Kenyan
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PROFESSIONAL PROFILE

A motivated Computer Science graduate (2026) with hands-on experience in ICT support, networking, and system installation gained during industrial attachment at Kenya Revenue Authority (KRA).

Built 4 real IoT projects including hardware-based sensor networks, REST APIs for device management, network intrusion detection, and containerized IoT platforms. Strong understanding of MQTT protocol, embedded systems, network security, and cloud integration.

Passionate about supporting the design, deployment, and maintenance of IoT and Smart City systems at Konza Technopolis. Ready to contribute to sensor installation, network configuration, technical documentation, and stakeholder coordination.

EDUCATION

Degree	Institution	Expected Graduation
Bachelor of Science in Computer Science	University of Embu	2026
Relevant Coursework: Computer Networks, Distributed Systems, Information Systems, Machine Learning, ICT Project Management		

PROFESSIONAL EXPERIENCE

ICT Intern (Industrial Attachment)
Kenya Revenue Authority (KRA) – ICT Department
2025 – 2026

Responsibility	Details
User Support	Provided end-user ICT support including troubleshooting hardware and software issues.
System Deployment	Installed and configured operating systems (Windows/Linux) and business applications.
Network Support	Assisted in structured cabling and basic network troubleshooting (TCP/IP, connectivity).
Desktop Management	Performed desktop deployment and system configuration for new and existing users.
Server Assistance	Assisted in server-related tasks (monitoring, basic maintenance) under senior supervision.
Documentation	Documented technical issues, resolutions, and provided user support feedback.

Skills Demonstrated: Network fundamentals, system installation, ICT service desk operations, technical documentation, hardware configuration, client communication

IoT PROJECTS

Project 1: Real MQTT Sensor Network with ESP8266

What I built: Programmed an ESP8266 microcontroller with a DHT11 temperature/humidity sensor to collect real environmental data and publish to cloud MQTT broker.

Technical details:

Embedded C++ firmware on ESP8266 (Arduino IDE)

MQTT protocol using HiveMQ public broker

Python subscriber for data ingestion and storage

Skills demonstrated: Embedded programming, hardware integration, MQTT protocol, sensor calibration, serial communication

Real-world application: Sensor data collection for smart city environmental monitoring

Project 2: IoT Device Management REST API

What I built: A REST API backend for registering, monitoring, and controlling IoT devices remotely (similar to Azure IoT Hub).

Technical details:

FastAPI framework (Python) for REST endpoints

SQLite database for device registry and telemetry storage

Device health tracking and last-seen monitoring

Skills demonstrated: REST API design, database modeling, CRUD operations, backend architecture

Real-world application: Device lifecycle management for large-scale IoT deployments

Project 3: Network Intrusion Detection System (IDS)

What I built: A packet sniffer that analyzes network traffic to detect port scans and SSH brute force attacks in real-time.

Technical details:

Scapy library for packet capture and analysis

Defaultdict for tracking connection attempts

Automatic IP blocking and alert generation

Skills demonstrated: Network security, packet analysis, anomaly detection, real-time monitoring

Real-world application: Cybersecurity for IoT networks and smart city infrastructure

Project 4: Dockerized IoT Stack (Full Platform)

What I built: Containerized an entire IoT platform including MQTT broker, time-series database, and visualization dashboard.

Technical details:

Docker Compose orchestrating 4 services: Mosquitto (MQTT), InfluxDB (database), Grafana (dashboard), Python consumer

Container networking and volume management

One-command deployment (docker-compose up)

Skills demonstrated: Containerization, orchestration, infrastructure as code, DevOps practices

Real-world application: Scalable IoT platform deployment for smart city operations

Other projects

Cross-Platform Firewall System

- Developed a firewall system focused on cross-platform security and network protection concepts
- Implemented rule-based traffic filtering logic to simulate cybersecurity enforcement mechanisms
- Demonstrates understanding of network security principles relevant to enterprise and IoT environments
- GitHub: <https://github.com/Rotichpatrick792/cross-platform-firewall>

Fault Detection System

- Designed a system for detecting faults in devices or systems using logical condition monitoring

- Focused on identifying anomalies and triggering alerts based on predefined system behaviours
- Demonstrates core IoT concept understanding in monitoring, automation, and system reliability
- GitHub: <https://github.com/Rotichpatrick792/fault-detector>

Afya Bora Health System

- Developed a healthcare-related system aimed at improving access to basic health services and information management
- Focused on structured data handling and user-centric system design
- Demonstrates ability to build real-world impact systems in public service domains
- GitHub: <https://github.com/Rotichpatrick792/afya-bora>

Job Connector AI (Career Matching System)

- Built a system to connect users with job opportunities using structured matching logic
- Focused on backend logic and data-driven decision-making workflows
- Demonstrates system design and intelligent application architecture
- GitHub: <https://github.com/Rotichpatrick792/job-connector>

Point of Sale (POS) System – Full Stack Development

- Developed a full-stack POS system with backend and frontend components
- Implemented sales processing, system logic, and user interface modules
- Demonstrates ability to build scalable business applications
- GitHub:
 - Backend: <https://github.com/Rotichpatrick792/pos-backend>
 - Frontend: <https://github.com/Rotichpatrick792/pos-frontend>

TECHNICAL SKILLS

Category	Skills	Proficiency
Networking	TCP/IP, LAN/WAN, Packet analysis, Port scanning	Intermediate
IoT Protocols	MQTT (publish/subscribe)	Working knowledge
Embedded Systems	ESP8266, Arduino IDE, DHT11 sensors	Project-level
Backend Development	FastAPI, REST APIs, SQLite	Intermediate
Security	Intrusion detection, Scapy, Network monitoring	Basic-Intermediate
DevOps	Docker, Docker Compose, Containerization	Working knowledge
Programming	Python, C++ (Arduino), SQL	Intermediate
Tools	Git/GitHub, Cisco Packet Tracer, Kali Linux, VS Code	Working knowledge

CERTIFICATIONS & TRAINING

Currently pursuing: Cisco Introduction to IoT

ADDITIONAL INFORMATION

Languages	English (Fluent), Kiswahili (Fluent)
Availability	Available full time
Work authorization	Kenyan citizen
Driving license	Yes

REFERENCES

1. KRA Supervisor – Steve Gitau (0726321415)
2. Uoem lec- Dr Faith(0737766052)